2.01 Complex Numbers: De Moivre's Theorem

Worked Solutions

1 a)

$$\begin{aligned}\left(\cos\frac{\pi}{6}+\mathrm{i}\sin\frac{\pi}{6}\right)^{7} &= \cos\frac{7\pi}{6}+\mathrm{i}\sin\frac{7\pi}{6}\\ &= \cos(-\frac{5\pi}{6})+\mathrm{i}\sin(-\frac{5\pi}{6})\end{aligned}$$

1 b)

$$\begin{aligned}\left(\cos\frac{2\pi}{3}+\mathrm{i}\sin\frac{2\pi}{3}\right)^{5} &= \cos\frac{10\pi}{3}+\mathrm{i}\sin\frac{10\pi}{3}\\ &= \cos(-\frac{2\pi}{3})+\mathrm{i}\sin(-\frac{2\pi}{3})\end{aligned}$$

1 c)

$$\begin{aligned}[2\left(\cos\frac{\pi}{5}+\mathrm{i}\sin\frac{\pi}{5}\right)]^{11} &= 2^{11}\left(\cos\frac{11\pi}{5}+\mathrm{i}\sin\frac{11\pi}{5}\right)\\ &= 2048\left(\cos\frac{\pi}{5}+\mathrm{i}\sin\frac{\pi}{5}\right)\end{aligned}$$

1 d)

$$\begin{aligned}[3\left(\cos\frac{2\pi}{5}+\mathrm{i}\sin\frac{2\pi}{5}\right)]^{8} &= 3^{8}\left(\cos\frac{16\pi}{5}+\mathrm{i}\sin\frac{16\pi}{5}\right)\\ &= 6561\left(\cos(-\frac{4\pi}{5})+\mathrm{i}\sin(-\frac{4\pi}{5})\right)\end{aligned}$$

1 e)

$$\begin{aligned}\left(\cos\frac{\pi}{4}+\mathrm{i}\sin\frac{\pi}{4}\right)^{-7} &= \cos(-\frac{7\pi}{4})+\mathrm{i}\sin(-\frac{7\pi}{4})\\ &= \cos\frac{\pi}{4}+\mathrm{i}\sin\frac{\pi}{4}\end{aligned}$$

1 f)

$$\begin{aligned}[2\left(\cos\frac{4\pi}{9}+\mathrm{i}\sin\frac{4\pi}{9}\right)]^{-4} &= 2^{-4}\left(\cos(-\frac{16\pi}{9})+\mathrm{i}\sin(-\frac{16\pi}{9})\right)\\ &= \frac{1}{16}\left(\cos\frac{2\pi}{9}+\mathrm{i}\sin\frac{2\pi}{9}\right)\end{aligned}$$

2 a)

$$\begin{aligned}\left(\cos\frac{\pi}{4}+\mathrm{i}\sin\frac{\pi}{4}\right)^{15} &= \cos\frac{15\pi}{4}+\mathrm{i}\sin\frac{15\pi}{4}\\ &= \frac{\sqrt{2}}{2}-\frac{\sqrt{2}}{2}\mathrm{i}\end{aligned}$$

2 b)

$$\begin{aligned}\left(\cos\frac{2\pi}{3}+\mathrm{i}\sin\frac{2\pi}{3}\right)^{10} &= \cos\frac{20\pi}{3}+\mathrm{i}\sin\frac{20\pi}{3}\\ &= -\frac{1}{2}+\frac{\sqrt{3}}{2}\mathrm{i}\end{aligned}$$

2 c)

$$\begin{aligned}[4\left(\cos\frac{3\pi}{8}+\mathrm{i}\sin\frac{3\pi}{8}\right)]^6 &= 4^6\left(\cos\frac{18\pi}{8}+\mathrm{i}\sin\frac{18\pi}{8}\right)\\ &= 4096\left(\frac{\sqrt{2}}{2}+\frac{\sqrt{2}}{2}\mathrm{i}\right)\\ &= 2048\sqrt{2}+2048\sqrt{2}\mathrm{i}\end{aligned}$$

2 d)

$$\begin{aligned}[\mathrm{i}\left(\cos\frac{5\pi}{4}+\mathrm{i}\sin\frac{5\pi}{4}\right)]^5 &= \mathrm{i}^5\left(\cos\frac{25\pi}{4}+\mathrm{i}\sin\frac{25\pi}{4}\right)\\ &= \mathrm{i}\left(\frac{\sqrt{2}}{2}+\frac{\sqrt{2}}{2}\mathrm{i}\right)\\ &= -\frac{\sqrt{2}}{2}+\frac{\sqrt{2}}{2}\mathrm{i}\end{aligned}$$

2 e)

$$\begin{aligned}[-3\left(\cos\frac{\pi}{3}+\mathrm{i}\sin\frac{\pi}{3}\right)]^6 &= (-3)^6\left(\cos\frac{6\pi}{3}+\mathrm{i}\sin\frac{6\pi}{3}\right)\\ &= 729\end{aligned}$$

2 f)

$$\begin{aligned}[2\left(\cos\frac{5\pi}{6}+\mathrm{i}\sin\frac{5\pi}{6}\right)]^{-4} &= 2^{-4}\left(\cos(-\frac{10\pi}{3})+\mathrm{i}\sin(-\frac{10\pi}{3})\right)\\ &= -\frac{1}{32}+\frac{\sqrt{3}}{32}\mathrm{i}\end{aligned}$$

3 a)

$$\begin{aligned}\left(\cos\frac{\pi}{5}+\mathrm{i}\sin\frac{\pi}{5}\right)^7 &= \cos\frac{7\pi}{5}+\mathrm{i}\sin\frac{7\pi}{5}\\ &= \cos(-\frac{3\pi}{5})+\mathrm{i}\sin(-\frac{3\pi}{5})\\ &= \mathrm{e}^{-\frac{3\pi}{5}\mathrm{i}}\end{aligned}$$

3 b)

$$\begin{aligned}\left(\cos\frac{\pi}{3}+\mathrm{i}\sin\frac{\pi}{3}\right)^{10} &= \cos\frac{10\pi}{3}+\mathrm{i}\sin\frac{10\pi}{3}\\ &= \cos(-\frac{2\pi}{3})+\mathrm{i}\sin(-\frac{2\pi}{3})\\ &= \mathrm{e}^{-\frac{2\pi}{3}\mathrm{i}}\end{aligned}$$

3 c)

$$\begin{aligned}[3\left(\cos\frac{\pi}{2}+\mathrm{i}\sin\frac{\pi}{2}\right)]^5 &= 3^5\left(\cos\frac{5\pi}{2}+\mathrm{i}\sin\frac{5\pi}{2}\right)\\ &= 243\left(\cos\frac{\pi}{2}+\mathrm{i}\sin\frac{\pi}{2}\right)\\ &= 243\mathrm{e}^{\frac{\pi}{2}\mathrm{i}}\end{aligned}$$

3 d)

$$\begin{aligned}[3\left(\cos\frac{2\pi}{5}+\mathrm{i}\sin\frac{2\pi}{5}\right)]^8 &= 3^8\left(\cos\frac{16\pi}{5}+\mathrm{i}\sin\frac{16\pi}{5}\right)\\ &= 6561\left(\cos(-\frac{4\pi}{5})+\mathrm{i}\sin(-\frac{4\pi}{5})\right)\\ &= 6561\mathrm{e}^{-\frac{4\pi}{5}\mathrm{i}}\end{aligned}$$

3 e)

$$\begin{aligned}[2\left(\cos\frac{\pi}{6}+\mathrm{i}\sin\frac{\pi}{6}\right)]^{-4} &= 2^{-4}\left(\cos(-\frac{4\pi}{6})+\mathrm{i}\sin(-\frac{4\pi}{6})\right)\\ &= \frac{1}{16}\left(\cos(-\frac{2\pi}{3})+\mathrm{i}\sin(-\frac{2\pi}{3})\right)\\ &= \frac{1}{16}\mathrm{e}^{-\frac{2\pi}{3}\mathrm{i}}\end{aligned}$$

3 f)

$$\begin{aligned}[-3\left(\cos\frac{5\pi}{8}+\mathrm{i}\sin\frac{5\pi}{8}\right)]^{-3} &= (-3)^{-3}\left(\cos(-\frac{15\pi}{8})+\mathrm{i}\sin(-\frac{15\pi}{8})\right)\\ &= -\frac{1}{27}\left(\cos\frac{\pi}{8}+\mathrm{i}\sin\frac{\pi}{8}\right)\\ &= \frac{1}{27}\left(\cos(-\frac{7\pi}{8})+\mathrm{i}\sin(-\frac{7\pi}{8})\right)\\ &= \frac{1}{27}\mathrm{e}^{-\frac{7\pi}{8}\mathrm{i}}\end{aligned}$$

4 a)

Let $z = 1+\sqrt{3}\mathrm{i}$ so that $|z| = \sqrt{1^2+3} = 2$ and $\arg z = \arctan\left(\frac{\sqrt{3}}{1}\right) = \frac{\pi}{3}$.

$$\begin{aligned}z &= 2\left(\cos\frac{\pi}{3}+\mathrm{i}\sin\frac{\pi}{3}\right)\ \text{So}\ (1+\sqrt{3}\mathrm{i})^7\\ &= [2\left(\cos\frac{\pi}{3}+\mathrm{i}\sin\frac{\pi}{3}\right)]^7\\ &= 2^7\left(\cos\frac{7\pi}{3}+\mathrm{i}\sin\frac{7\pi}{3}\right)\\ &= 128\left(\cos\frac{\pi}{3}+\mathrm{i}\sin\frac{\pi}{3}\right)\end{aligned}$$

4 b)

Let $z = 1-\mathrm{i}$ so that $|z| = \sqrt{1^2+(-1)^2} = \sqrt{2}$ and $\arg z = -\arctan\left(\frac{1}{1}\right) = -\frac{\pi}{4}$.

$$\begin{aligned}z &= \sqrt{2}\left(\cos(-\frac{\pi}{4})+\mathrm{i}\sin(-\frac{\pi}{4})\right)\ \text{So}\ (1-\mathrm{i})^{10}\\ &= [\sqrt{2}\left(\cos(-\frac{\pi}{4})+\mathrm{i}\sin(-\frac{\pi}{4})\right)]^{10}\\ &= (\sqrt{2})^{10}\left(\cos(-\frac{10\pi}{4})+\mathrm{i}\sin(-\frac{10\pi}{4})\right)\\ &= 32\left(\cos(-\frac{\pi}{2})+\mathrm{i}\sin(-\frac{\pi}{2})\right)\end{aligned}$$

4 c)

Let $z = 2\mathrm{i}$ so that $|z| = 2$ and $\arg z = \frac{\pi}{2}$.

$$\begin{aligned}z &= 2\left(\cos\frac{\pi}{2}+\mathrm{i}\sin\frac{\pi}{2}\right)\ \text{So}\ (2\mathrm{i})^{20}\\ &= [2\left(\cos\frac{\pi}{2}+\mathrm{i}\sin\frac{\pi}{2}\right)]^{20}\\ &= 2^{20}\left(\cos\frac{20\pi}{2}+\mathrm{i}\sin\frac{20\pi}{2}\right)\\ &= 1048576\left(\cos 0+\mathrm{i}\sin 0\right)\end{aligned}$$

4 d)

Let $z = -8 + 8\sqrt{3}\mathrm{i}$ so that $|z| = \sqrt{(-8)^2 + (8\sqrt{3})^2} = 16$ and $\arg z = \pi - \arctan\left(\frac{8\sqrt{3}}{8}\right) = \frac{2\pi}{3}$.

$$\begin{aligned} z &= 16\left(\cos\frac{2\pi}{3} + \mathrm{i}\sin\frac{2\pi}{3}\right) \text{ So } (-8 + 8\sqrt{3}\mathrm{i})^5 \\ &= [16\left(\cos\frac{2\pi}{3} + \mathrm{i}\sin\frac{2\pi}{3}\right)]^5 \\ &= 16^5\left(\cos\frac{10\pi}{3} + \mathrm{i}\sin\frac{10\pi}{3}\right) \\ &= 1048576\left(\cos\frac{10\pi}{3} + \mathrm{i}\sin\frac{10\pi}{3}\right) \\ &= 1048576\left(\cos(-\frac{2\pi}{3}) + \mathrm{i}\sin(-\frac{2\pi}{3})\right) \end{aligned}$$

5)

Let $z = -1 + \sqrt{3}\mathrm{i}$ so that $|z| = \sqrt{(-1)^2 + 3} = 2$ and $\arg z = \pi - \arctan\left(\frac{\sqrt{3}}{1}\right) = \frac{2\pi}{3}$.

$$\begin{aligned} z &= 2\left(\cos\frac{2\pi}{3} + \mathrm{i}\sin\frac{2\pi}{3}\right) \text{ So } (-1 + \sqrt{3}\mathrm{i})^{-4} \\ &= [2\left(\cos\frac{2\pi}{3} + \mathrm{i}\sin\frac{2\pi}{3}\right)]^{-4} \\ &= 2^{-4}\left(\cos(-\frac{8\pi}{3}) + \mathrm{i}\sin(-\frac{8\pi}{3})\right) \\ &= -\frac{1}{32} - \frac{\sqrt{3}}{32}\mathrm{i} \end{aligned}$$

6)

Let $z = 2 - 2\mathrm{i}$ so that $|z| = \sqrt{2^2 + (-2)^2} = 2\sqrt{2}$ and $\arg z = -\arctan\left(\frac{2}{2}\right) = -\frac{\pi}{4}$.

$$\begin{aligned} z &= 2\sqrt{2}\left(\cos(-\frac{\pi}{4}) + \mathrm{i}\sin(-\frac{\pi}{4})\right) \text{ So } (2 - 2\mathrm{i})^{-5} \\ &= [2\sqrt{2}\left(\cos(-\frac{\pi}{4}) + \mathrm{i}\sin(-\frac{\pi}{4})\right)]^{-5} \\ &= (2\sqrt{2})^{-5}\left(\cos\frac{5\pi}{4} + \mathrm{i}\sin\frac{5\pi}{4}\right) \\ &= -\frac{1}{256} - \frac{1}{256}\mathrm{i} \end{aligned}$$

7 a)

$$\cos\theta - \mathrm{i}\sin\theta = \cos(-\theta) + \mathrm{i}\sin(-\theta)$$

7 b)

$$(\cos(-\theta) + \mathrm{i}\sin(-\theta))^8 = \cos(-8\theta) + \mathrm{i}\sin(-8\theta)$$

8)

$$\begin{aligned}\cos\frac{\pi}{5}-\mathrm{i}\sin\frac{\pi}{5}&=\cos(-\frac{\pi}{5})+\mathrm{i}\sin(-\frac{\pi}{5})\text{ So }\left(\cos(-\frac{\pi}{5})+\mathrm{i}\sin(-\frac{\pi}{5})\right)^9\\&=\cos(-\frac{9\pi}{5})+\mathrm{i}\sin(-\frac{9\pi}{5})\\&=\cos\frac{\pi}{5}+\mathrm{i}\sin\frac{\pi}{5}\end{aligned}$$

9 a)

Let $z=1+\mathrm{i}$ so that $|z|=\sqrt{1^2+1^2}=\sqrt{2}$ and $\arg z=\arctan\left(\frac{1}{1}\right)=\frac{\pi}{4}$.

$$\begin{aligned}z&=\sqrt{2}\left(\cos\frac{\pi}{4}+\mathrm{i}\sin\frac{\pi}{4}\right)\\\text{So }(1+\mathrm{i})^8&=[\sqrt{2}\left(\cos\frac{\pi}{4}+\mathrm{i}\sin\frac{\pi}{4}\right)]^8\\&=(\sqrt{2})^8\left(\cos\frac{8\pi}{4}+\mathrm{i}\sin\frac{8\pi}{4}\right)\\&=16\end{aligned}$$

Let $z=1-\mathrm{i}$ so that $|z|=\sqrt{1^2+(-1)^2}=\sqrt{2}$ and $\arg z=-\arctan\left(\frac{1}{1}\right)=-\frac{\pi}{4}$.

$$\begin{aligned}z&=\sqrt{2}\left(\cos(-\frac{\pi}{4})+\mathrm{i}\sin(-\frac{\pi}{4})\right)\\\text{So }(1-\mathrm{i})^8&=[\sqrt{2}\left(\cos(-\frac{\pi}{4})+\mathrm{i}\sin(-\frac{\pi}{4})\right)]^8\\&=(\sqrt{2})^8\left(\cos(-\frac{8\pi}{4})+\mathrm{i}\sin(-\frac{8\pi}{4})\right)\\&=16\end{aligned}$$

$$\therefore(1+\mathrm{i})^8+(1-\mathrm{i})^8=16+16=32$$

9 b)

Let $z=\sqrt{3}+\mathrm{i}$ so that $|z|=\sqrt{3+1^2}=2$ and $\arg z=\arctan\left(\frac{1}{\sqrt{3}}\right)=\frac{\pi}{6}$.

$$\begin{aligned}z&=2\left(\cos\frac{\pi}{6}+\mathrm{i}\sin\frac{\pi}{6}\right)\\\text{So }(\sqrt{3}+\mathrm{i})^7&=[2\left(\cos\frac{\pi}{6}+\mathrm{i}\sin\frac{\pi}{6}\right)]^7\\&=2^7\left(\cos\frac{7\pi}{6}+\mathrm{i}\sin\frac{7\pi}{6}\right)\\&=-64\sqrt{3}-64\mathrm{i}\end{aligned}$$

Let $z=\sqrt{3}-\mathrm{i}$ so that $|z|=\sqrt{3+(-1)^2}=2$ and $\arg z=-\arctan\left(\frac{1}{\sqrt{3}}\right)=-\frac{\pi}{6}$.

$$z=2\left(\cos(-\frac{\pi}{6})+\mathrm{i}\sin(-\frac{\pi}{6})\right)$$

$$\begin{aligned}\text{So } (\sqrt{3}-\mathrm{i})^7 &= [2\left(\cos(-\frac{\pi}{6})+\mathrm{i}\sin(-\frac{\pi}{6})\right)]^7\\ &= 2^7\left(\cos(-\frac{7\pi}{6})+\mathrm{i}\sin(-\frac{7\pi}{6})\right)\\ &= -64\sqrt{3}+64\mathrm{i}\end{aligned}$$

$$\therefore (\sqrt{3}+\mathrm{i})^7+(\sqrt{3}-\mathrm{i})^7 = -64\sqrt{3}-64\mathrm{i}+-64\sqrt{3}+64\mathrm{i} = -128\sqrt{3}$$

10)

$$\begin{aligned}(\cos 9\theta+\mathrm{i}\sin 9\theta)(\cos 6\theta+\mathrm{i}\sin 6\theta) &= (\cos\theta+\mathrm{i}\sin\theta)^9(\cos\theta+\mathrm{i}\sin\theta)^6\\ &= (\cos\theta+\mathrm{i}\sin\theta)^{15}\\ &= \cos 15\theta+\mathrm{i}\sin 15\theta\end{aligned}$$

11 a)

$$\begin{aligned}\left(\cos\frac{4\pi}{3}+\mathrm{i}\sin\frac{4\pi}{3}\right)\left(\cos\frac{7\pi}{3}+\mathrm{i}\sin\frac{7\pi}{3}\right) &= \left(\cos\frac{\pi}{3}+\mathrm{i}\sin\frac{\pi}{3}\right)^4\left(\cos\frac{\pi}{3}+\mathrm{i}\sin\frac{\pi}{3}\right)^7\\ &= \left(\cos\frac{\pi}{3}+\mathrm{i}\sin\frac{\pi}{3}\right)^{11}\\ &= \cos\frac{11\pi}{3}+\mathrm{i}\sin\frac{11\pi}{3}\\ &= \cos(-\frac{\pi}{3})+\mathrm{i}\sin(-\frac{\pi}{3})\end{aligned}$$

11 b)

$$\begin{aligned}\left(\cos\frac{17\pi}{4}+\mathrm{i}\sin\frac{17\pi}{4}\right)\left(\cos\frac{5\pi}{4}+\mathrm{i}\sin\frac{5\pi}{4}\right) &= \left(\cos\frac{\pi}{4}+\mathrm{i}\sin\frac{\pi}{4}\right)^{17}\left(\cos\frac{\pi}{4}+\mathrm{i}\sin\frac{\pi}{4}\right)^5\\ &= \left(\cos\frac{\pi}{4}+\mathrm{i}\sin\frac{\pi}{4}\right)^{22}\\ &= \cos\frac{22\pi}{4}+\mathrm{i}\sin\frac{22\pi}{4}\\ &= \cos(-\frac{\pi}{2})+\mathrm{i}\sin(-\frac{\pi}{2})\end{aligned}$$

11 c)

$$\begin{aligned}\left(\cos\frac{7\pi}{2}+\mathrm{i}\sin\frac{7\pi}{2}\right)\left(\cos\frac{13\pi}{2}-\mathrm{i}\sin\frac{13\pi}{2}\right) &= \left(\cos\frac{7\pi}{2}+\mathrm{i}\sin\frac{7\pi}{2}\right)\left(\cos(-\frac{13\pi}{2})-\mathrm{i}\sin(-\frac{13\pi}{2})\right)\\ &= \left(\cos\frac{\pi}{2}+\mathrm{i}\sin\frac{\pi}{2}\right)^7\left(\cos\frac{\pi}{2}+\mathrm{i}\sin\frac{\pi}{2}\right)^{-13}\\ &= \left(\cos\frac{\pi}{2}+\mathrm{i}\sin\frac{\pi}{2}\right)^{-6}\\ &= \cos(-\frac{6\pi}{2})+\mathrm{i}\sin(-\frac{6\pi}{2})\\ &= \cos\pi+\mathrm{i}\sin\pi\end{aligned}$$

12 a)

$$\begin{aligned}\left(\cos\frac{\pi}{6}+\mathrm{i}\sin\frac{\pi}{6}\right)^4\left(\cos\frac{\pi}{3}-\mathrm{i}\sin\frac{\pi}{3}\right)^5 &= \left(\cos\frac{4\pi}{6}+\mathrm{i}\sin\frac{4\pi}{6}\right)\left(\cos(-\frac{\pi}{3})+\mathrm{i}\sin(-\frac{\pi}{3})\right)^5\\ &= \left(\cos\frac{2\pi}{3}+\mathrm{i}\sin\frac{2\pi}{3}\right)\left(\cos(-\frac{5\pi}{3})+\mathrm{i}\sin(-\frac{5\pi}{3})\right)\\ &= \cos(-\pi)+\mathrm{i}\sin(-\pi)\end{aligned}$$

$$\therefore z = \cos\pi+\mathrm{i}\sin\pi$$

12 b)

$$z=-1$$

13 a)

$$\begin{aligned}\frac{(\cos 9\theta+\mathrm{i}\sin 9\theta)^4}{(\cos 4\theta+\mathrm{i}\sin 4\theta)^6} &= \frac{\cos 36\theta+\mathrm{i}\sin 36\theta}{\cos 24\theta+\mathrm{i}\sin 24\theta}\\ &= \cos(36\theta-24\theta)+\mathrm{i}\sin(36\theta-24\theta)\\ &= \cos 12\theta+\mathrm{i}\sin 12\theta\end{aligned}$$

13 b)

$$\begin{aligned}\frac{(\cos 5\theta+\mathrm{i}\sin 5\theta)^8}{[2(\cos 3\theta+\mathrm{i}\sin 3\theta)]^6} &= \frac{\cos 40\theta+\mathrm{i}\sin 40\theta}{64(\cos 18\theta+\mathrm{i}\sin 18\theta)}\\ &= \frac{1}{64}\left(\cos(40\theta-18\theta)+\mathrm{i}\sin(40\theta-18\theta)\right)\\ &= \frac{1}{64}\left(\cos 22\theta+\mathrm{i}\sin 22\theta\right)\end{aligned}$$

14 a)

$$\begin{aligned}\frac{\cos\frac{4\pi}{11}+\mathrm{i}\sin\frac{4\pi}{11}}{\cos\frac{5\pi}{11}-\mathrm{i}\sin\frac{5\pi}{11}} &= \frac{\cos\frac{4\pi}{11}+\mathrm{i}\sin\frac{4\pi}{11}}{\cos(-\frac{5\pi}{11})+\mathrm{i}\sin(-\frac{5\pi}{11})}\\ &= \cos(\frac{4\pi}{11}--\frac{5\pi}{11})+\mathrm{i}\sin(\frac{4\pi}{11}--\frac{5\pi}{11})\\ &= \cos\frac{9\pi}{11}+\mathrm{i}\sin\frac{9\pi}{11}\end{aligned}$$

14 b)

$$\begin{aligned}
\frac{\left(\cos\frac{9\pi}{14}+\mathrm{i}\sin\frac{9\pi}{14}\right)^6}{\left(\cos\frac{3\pi}{14}-\mathrm{i}\sin\frac{3\pi}{14}\right)^4} &= \frac{\left(\cos\frac{9\pi}{14}+\mathrm{i}\sin\frac{9\pi}{14}\right)^6}{\left(\cos(-\frac{3\pi}{14})+\mathrm{i}(-\frac{3\pi}{14})\right)^4} \\
&= \frac{\cos\frac{54\pi}{14}+\mathrm{i}\sin\frac{54\pi}{14}}{\cos(-\frac{12\pi}{14})+\mathrm{i}(-\frac{12\pi}{14})} \\
&= \cos(\frac{54\pi}{14}--\frac{12\pi}{14})+\mathrm{i}\sin(\frac{54\pi}{14}--\frac{12\pi}{14}) \\
&= \cos\frac{33\pi}{7}+\mathrm{i}\sin\frac{33\pi}{7} \\
&= \cos\frac{5\pi}{7}+\mathrm{i}\sin\frac{5\pi}{7}
\end{aligned}$$

15)

Let $z = a + b\mathrm{i} = r^r(\cos\theta + \mathrm{i}\sin\theta)$

$$\begin{aligned}
\text{So that } (a+b\mathrm{i})^n &= r^n(\cos\theta+\mathrm{i}\sin\theta)^n \\
&= r^n(\cos n\theta+\mathrm{i}\sin n\theta) \\
\text{And } (a-b\mathrm{i})^n &= r^n(\cos\theta-\mathrm{i}\sin\theta)^n \\
&= r^n(\cos(-\theta)+\mathrm{i}\sin(-\theta))^n \\
&= r^n(\cos(-n\theta)+\mathrm{i}\sin(-n\theta)) \\
\therefore (a+b\mathrm{i})^n+(a-b\mathrm{i})^n &= r^n(\cos n\theta+\mathrm{i}\sin n\theta+\cos(-n\theta)+\mathrm{i}\sin(-n\theta)) \\
&= r^n(\cos n\theta+\mathrm{i}\sin n\theta+\cos n\theta-\mathrm{i}\sin n\theta) \\
&= 2r^n\cos n\theta \quad \text{which is real.}
\end{aligned}$$

16)

$\sin\pi = 0$ so since $\left(\cos\frac{\pi}{6}+\mathrm{i}\sin\frac{\pi}{6}\right)^6 = \cos\pi+\mathrm{i}\sin\pi$

$$n = 6$$

17 a)

$$\left(\cos\frac{4\pi}{5}+\mathrm{i}\sin\frac{4\pi}{5}\right)^n = \cos\frac{4n\pi}{5}+\mathrm{i}\sin\frac{4n\pi}{5}$$

$\sin\frac{4n\pi}{5} = 0$ so $\frac{4n\pi}{5} = k\pi$ for any integer k

$n = \frac{5}{4}k$ so choose $k = 4$

$$\therefore n = 5$$

17 b)

$$z = 3^5\cos 4\pi = 243$$

18 a)

Let $z_1 = \sqrt{3} - \mathrm{i}$ so that $|z_1| = \sqrt{3 + (-1)^2} = 2$ and $\arg z_1 = -\arctan\left(\frac{1}{\sqrt{3}}\right) = -\frac{\pi}{6}$

$$z_1 = 2\left(\cos\left(-\frac{\pi}{6}\right) + \mathrm{i}\sin\left(-\frac{\pi}{6}\right)\right)$$

Let $z_2 = \sqrt{3} + \mathrm{i}$ so that $|z_2| = \sqrt{3 + 1^2} = 2$ and $\arg z_2 = \arctan\left(\frac{1}{\sqrt{3}}\right) = \frac{\pi}{6}$

$$z_2 = 2\left(\cos\frac{\pi}{6} + \mathrm{i}\sin\frac{\pi}{6}\right)$$

$$z = \frac{z_1}{z_2} = \frac{2\left(\cos\left(-\frac{\pi}{6}\right) + \mathrm{i}\sin\left(-\frac{\pi}{6}\right)\right)}{2\left(\cos\frac{\pi}{6} + \mathrm{i}\sin\frac{\pi}{6}\right)} = \cos\left(-\frac{\pi}{3}\right) + \mathrm{i}\sin\left(-\frac{\pi}{3}\right)$$

18 b)

$$z^n = \left[\cos\left(-\frac{\pi}{3}\right) + \mathrm{i}\sin\left(-\frac{\pi}{3}\right)\right]^n$$

$\sin\left(-\frac{n\pi}{3}\right) = 0$ so $-\frac{n\pi}{3} = k\pi$ for any integer k

$n = -3k$ so choose $k = -1$

$$\therefore n = 3$$

19 a)

$|z| = \sqrt{2^2 + 2^2} = 2\sqrt{2}$ and $\arg z = \arctan\left(\frac{2}{2}\right) = \frac{\pi}{4}$

$$z = 2\sqrt{2}\left(\cos\frac{\pi}{4} + \mathrm{i}\sin\frac{\pi}{4}\right)$$

19 b)

$$z^n = \left[2\sqrt{2}\left(\cos\frac{\pi}{4} + \mathrm{i}\sin\frac{\pi}{4}\right)\right]^n$$

$\cos\frac{n\pi}{4} = 0$ so $\frac{n\pi}{4} = k\pi + \frac{\pi}{2}$ for any integer k

$n = 4k + 2$ so choose $k = 0$

$$\therefore n = 2$$

20 a)

$$\frac{\cos\frac{m\pi}{6} + \mathrm{i}\sin\frac{m\pi}{6}}{\cos\frac{n\pi}{4} + \mathrm{i}\sin\frac{n\pi}{4}} = \cos\left(\frac{m\pi}{6} - \frac{n\pi}{4}\right) + \mathrm{i}\sin\left(\frac{m\pi}{6} - \frac{n\pi}{4}\right)$$

$\cos\left(\frac{m\pi}{6} - \frac{n\pi}{4}\right) = 0$ so $\frac{m\pi}{6} - \frac{n\pi}{4} = k\pi + \frac{\pi}{2}$ for any integer k

$\frac{m}{6} - \frac{n}{4} = k + \frac{1}{2} \Rightarrow 2m - 3n = 12k + 6$ so choose $k = 0$

$$2m - 3n = 6$$

When $n = 2$, $m = 6$

$$\therefore n = 2$$

20 b)

For z to be real, $\sin\theta = 0$ so $\theta = k\pi$ for some integer k

$$m = \frac{3n}{2} + 6k$$

n must be even, so that $n = 2$. With $k = 0$ then $m = 3$

So $z = 1$

21 a)

$$\alpha\beta\gamma = 4, \quad \alpha\beta + \alpha\gamma + \beta\gamma = q, \quad \alpha + \beta + \gamma = p$$

Since $1 + \mathrm{i}$ is a root, then $1 - \mathrm{i}$ is also a root.

So $(1 + \mathrm{i})(1 - \mathrm{i})\gamma = 4 \Rightarrow 2\gamma = 4$

$$p = 1 + \mathrm{i} + 1 - \mathrm{i} + 2 = 4$$

and $q = (1 + \mathrm{i})(1 - \mathrm{i}) + 2(1 + \mathrm{i}) + 2(1 - \mathrm{i}) = 6$

21 b)

The three roots are $1 + \mathrm{i}$, $1 - \mathrm{i}$ and 2

Let $z = 1 + \mathrm{i}$ so that $|z| = \sqrt{1^2 + 1^2} = \sqrt{2}$ and $\arg z = \arctan\left(\frac{1}{1}\right) = \frac{\pi}{4}$

$$z = \sqrt{2}\left(\cos\frac{\pi}{4} + \mathrm{i}\sin\frac{\pi}{4}\right)$$

$$\text{So } (1 + \mathrm{i})^n = \left[\sqrt{2}\left(\cos\frac{\pi}{4} + \mathrm{i}\sin\frac{\pi}{4}\right)\right]^n = (\sqrt{2})^n\left(\cos\frac{n\pi}{4} + \mathrm{i}\sin\frac{n\pi}{4}\right)$$

Let $z = 1 - \mathrm{i}$ so that $|z| = \sqrt{1^2 + (-1)^2} = \sqrt{2}$ and $\arg z = -\arctan\left(\frac{1}{1}\right) = -\frac{\pi}{4}$

$$z = \sqrt{2}\left(\cos\left(-\frac{\pi}{4}\right) + \mathrm{i}\sin\left(-\frac{\pi}{4}\right)\right)$$

$$\text{So } (1 - \mathrm{i})^n = \left[\sqrt{2}\left(\cos\left(-\frac{\pi}{4}\right) + \mathrm{i}\sin\left(-\frac{\pi}{4}\right)\right)\right]^n = (\sqrt{2})^n\left(\cos\left(-\frac{n\pi}{4}\right) + \mathrm{i}\sin\left(-\frac{n\pi}{4}\right)\right)$$

So

$$\begin{aligned}(1 + \mathrm{i})^n + (1 - \mathrm{i})^n + 2^n &= 2^{\frac{n}{2}}\left(\cos\frac{n\pi}{4} + \mathrm{i}\sin\frac{n\pi}{4}\right) + 2^{\frac{n}{2}}\left(\cos\left(-\frac{n\pi}{4}\right) + \mathrm{i}\sin\left(-\frac{n\pi}{4}\right)\right) + 2^n \\ &= 2^{\frac{n}{2}}\left(\cos\frac{n\pi}{4} + \mathrm{i}\sin\frac{n\pi}{4}\right) + 2^{\frac{n}{2}}\left(\cos\frac{n\pi}{4} - \mathrm{i}\sin\frac{n\pi}{4}\right) + 2^n \\ &= 2 \times 2^{\frac{n}{2}}\cos\frac{n\pi}{4} + 2^n \\ &= 2^{\frac{n}{2}+1}\cos\frac{n\pi}{4} + 2^n\end{aligned}$$

22 a)

$$\alpha\beta\gamma = q, \quad \alpha\beta + \alpha\gamma + \beta\gamma = 12, \quad \alpha + \beta + \gamma = p$$

Since $1 + \sqrt{3}\mathrm{i}$ is a root, then $1 - \sqrt{3}\mathrm{i}$ is also a root.

$$\begin{aligned}\text{So } (1 + \sqrt{3}\mathrm{i})(1 - \sqrt{3}\mathrm{i}) + (1 + \sqrt{3}\mathrm{i})\gamma + (1 - \sqrt{3}\mathrm{i})\gamma &= 13 \\ \Rightarrow 4 + 2\gamma &= 12 \Rightarrow \gamma \\ &= 4\end{aligned}$$

$p = 1 + \sqrt{3}\mathrm{i} + 1 - \sqrt{3}\mathrm{i} + 4 = 6$ and $q = 4(1 + \sqrt{3}\mathrm{i})(1 - \sqrt{3}\mathrm{i}) = 16$

22 b)

The three roots are $1+\sqrt{3}\mathrm{i}$, $1-\sqrt{3}\mathrm{i}$ and 4

Let $z = 1+\sqrt{3}\mathrm{i}$ so that $|z| = \sqrt{1^2+\sqrt{3}^2} = 2$ and $\arg z = \arctan\left(\frac{\sqrt{3}}{1}\right) = \frac{\pi}{3}$

$$z = 2\left(\cos\frac{\pi}{3} + \mathrm{i}\sin\frac{\pi}{3}\right)$$

$$\text{So } 1+\sqrt{3}\mathrm{i} = \left[2\left(\cos\frac{\pi}{3} + \mathrm{i}\sin\frac{\pi}{3}\right)\right]^n = 2^n\left(\cos\frac{n\pi}{3} + \mathrm{i}\sin\frac{n\pi}{3}\right)$$

Let $z = 1-\sqrt{3}\mathrm{i}$ so that $|z| = \sqrt{1^2+\sqrt{3}^2} = 2$ and $\arg z = -\arctan\left(\frac{\sqrt{3}}{1}\right) = -\frac{\pi}{3}$

$$z = 2\left(\cos\left(-\frac{\pi}{3}\right) + \mathrm{i}\sin\left(-\frac{\pi}{3}\right)\right)$$

$$\text{So } (1-\sqrt{3}\mathrm{i})^n = \left[2\left(\cos\left(-\frac{\pi}{3}\right) + \mathrm{i}\sin\left(-\frac{\pi}{3}\right)\right)\right]^n = 2^n\left(\cos\left(-\frac{n\pi}{3}\right) + \mathrm{i}\sin\left(-\frac{n\pi}{3}\right)\right)$$

So

$$\begin{aligned}(1+\sqrt{3}\mathrm{i})^n + (1-\sqrt{3}\mathrm{i})^n + 4^n &= 2^n\left(\cos\frac{n\pi}{3} + \mathrm{i}\sin\frac{n\pi}{3}\right) + 2^n\left(\cos\left(-\frac{n\pi}{3}\right) + \mathrm{i}\sin\left(-\frac{n\pi}{3}\right)\right) + 4^n \\ &= 2^n\left(\cos\frac{n\pi}{3} + \mathrm{i}\sin\frac{n\pi}{3}\right) + 2^n\left(\cos\frac{n\pi}{3} - \mathrm{i}\sin\frac{n\pi}{3}\right) + 4^n \\ &= 2\times 2^n\cos\frac{n\pi}{3} + 4^n \\ &= 2^{n+1}\cos\frac{n\pi}{3} + 2^{2n}\end{aligned}$$

23 a)

$$\begin{aligned}(\cosh z + \sinh z)^n &= \left[\frac{1}{2}(\mathrm{e}^z + \mathrm{e}^{-z}) + \frac{1}{2}(\mathrm{e}^z - \mathrm{e}^{-z})\right]^n \\ &= \left[\frac{1}{2}\mathrm{e}^z + \frac{1}{2}\mathrm{e}^z\right]^n \\ &= \mathrm{e}^{nz}\end{aligned}$$

But since $\cosh nz = \frac{1}{2}(\mathrm{e}^{nz} + \mathrm{e}^{-nz})$ and $\sinh z = \frac{1}{2}(\mathrm{e}^{nz} - \mathrm{e}^{-nz})$

$$\text{Then } \cosh nz + \sinh nz = \frac{1}{2}(\mathrm{e}^{nz} + \mathrm{e}^{-nz}) + \frac{1}{2}(\mathrm{e}^{nz} - \mathrm{e}^{-nz}) = \mathrm{e}^{nz}$$

$$\therefore (\cosh z + \sinh z)^n = \cosh nz + \sinh nz \quad \text{as req.}$$

23 b)

$$\begin{aligned}\cosh(\ln 2) &= \frac{1}{2}\left(2 + \frac{1}{2}\right) \\ &= \frac{5}{4} \\ \sinh(\ln 2) &= \frac{1}{2}\left(2 - \frac{1}{2}\right)\end{aligned}$$

$$
\begin{aligned}
&= \frac{3}{4} \\
\cosh(3\ln 2) + \sinh(3\ln 2) &= (\cosh(\ln 2) + \sinh(\ln 2))^3 \\
&= \left(\frac{5}{4} + \frac{3}{4}\right)^3 \\
&= 8
\end{aligned}
$$

Alternatively:

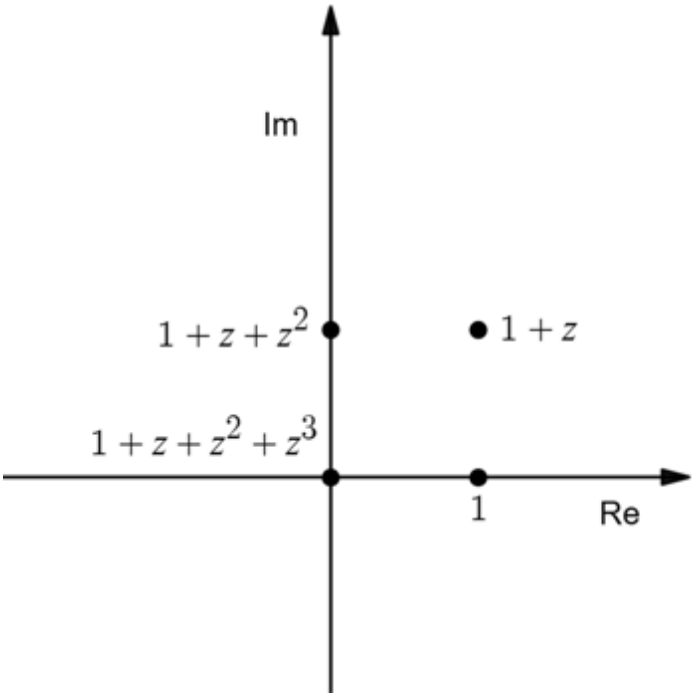
$$\cosh(3\ln 2) + \sinh(3\ln 2) = (\cosh(\ln 2) + \sinh(\ln 2))^3 = \mathrm{e}^{3\ln 2} = 8$$

24 a)

$$
\begin{aligned}
z^2 &= \left(\cos\frac{\pi}{2} + \mathrm{i}\sin\frac{\pi}{2}\right)^2 \\
&= \cos\pi + \mathrm{i}\sin\pi \\
z^3 &= \left(\cos\frac{\pi}{2} + \mathrm{i}\sin\frac{\pi}{2}\right)^3 \\
&= \cos\frac{3\pi}{2} + \mathrm{i}\sin\frac{3\pi}{2} \\
&= \cos\left(-\frac{\pi}{2}\right) + \mathrm{i}\sin\left(-\frac{\pi}{2}\right)
\end{aligned}
$$

24 b)

$$
\begin{aligned}
1 + z &= 1 + \cos\frac{\pi}{2} + \mathrm{i}\sin\frac{\pi}{2} \\
&= 1 + \mathrm{i} \\
1 + z + z^2 &= 1 + \mathrm{i} + \cos\pi + \mathrm{i}\sin\pi \\
&= \mathrm{i} \\
1 + z + z^2 + z^3 &= \mathrm{i} + \cos\left(-\frac{\pi}{2}\right) + \mathrm{i}\sin\left(-\frac{\pi}{2}\right) \\
&= 0
\end{aligned}
$$



24 c)

i) $\arg(1+z) = \frac{\pi}{4}$

ii) $\arg(1+z+z^2) = \frac{\pi}{2}$